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Tow less, dive smarter

A proposed manta tow and cull dive protocol

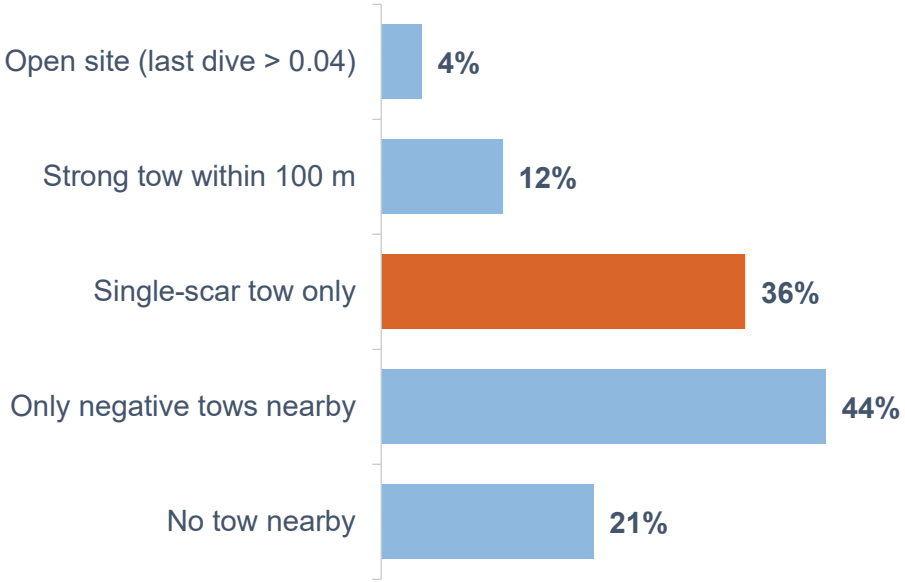
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For discussion with GBRMPA, November 2026



Most ghost dives follow weak evidence

Why was each site dived... and how often did it find nothing? 18,182 cull dives, 2019–2026



Share of dives that found no COTS (ghost dives). Dives: open 7,273; strong 3,752; single scar 5,393; negative only 1,037; no tow 612.

59%

of ghost dives followed a single-scar tow (1,952 of 3,308)

4–8 divers

in the water for every ghost dive

Towing is cheaper... but on active reefs much of it re-tows ground covered weeks earlier

Strong vs weak: write the trigger down

GBRMPA's scar analysis asked the right question... is one feeding scar enough to send divers in?

Strong

At least 1 COTS, or 2+ feeding scars, or scars 'common'

7.6%	12%	47%
of tows	ghost rate	> 0.04

Weak

No COTS, and exactly 1 scar or scars 'present'

7.2%	36%	16%
of tows	ghost rate	> 0.04

Negative

No COTS and no feeding scars

85%	44%	16%
of tows	ghost rate	> 0.04

Raise the trigger to two scars? Skipping every weak-scar dive avoids 59% of ghost dives... but misses 11% of dives above 0.04 and forgoes 10% of COTS culled. **Weak evidence isn't worthless: 16% of weak-scar dives were above 0.04.**

The definitions alone are still worth adopting... they make triggers consistent between crews.

The site priority ranking: A to E, within every reef

Habitat + the last 12 months of dives and tows... a useful nudge, not a prediction

1. **Habitat map**
   XGBoost on past cull dives at 30 m: seabed shape, coral habitat, waves

2. **Recent evidence**
   Last 12 months: problematic dives and strong tows nearby

3. **Scoring**
   Refitted each season on earlier seasons only

4. **Classes**
   Ranked within the reef, refreshed before each voyage

A	B	C	D	E
top 20%	to 50%	to 66%	to 80%	last 20%



AUC 0.64 forward in time (0.58–0.74 by season); habitat alone 0.54–0.58

The protocol: six steps, three of them new

Strong evidence still drives the dive... the ranking only decides the rest

1 NEW

Refresh the ranking

Before each voyage: maps with A–E classes and the tow zone

2

Dive open sites first

Keep culling until CPUE drops below 0.04

3 NEW

Tow

First visit: whole reef. Follow-up visit: only the zone (A–B, open, flagged earlier)

4

Dive strong evidence

Every site within 100 m of a strong tow, whatever its class

5 NEW

The rest: A–C only

Weak scar, negative tows or a crew call... only at A–C sites

6

Crews can override

COTS or fresh feeding seen? Dive, and record why

What it would have done, 2019–2026

Replayed on eight seasons of records... using only what crews knew at the time

30%

of ghost dives avoided

~124 a season

19%

of low-density dives avoided

~97 a season; 13% of bottom time

2.8%

of dives above 0.04 missed

~28 a season; mean CPUE 0.099, just over target

20%

of follow-up tows saved

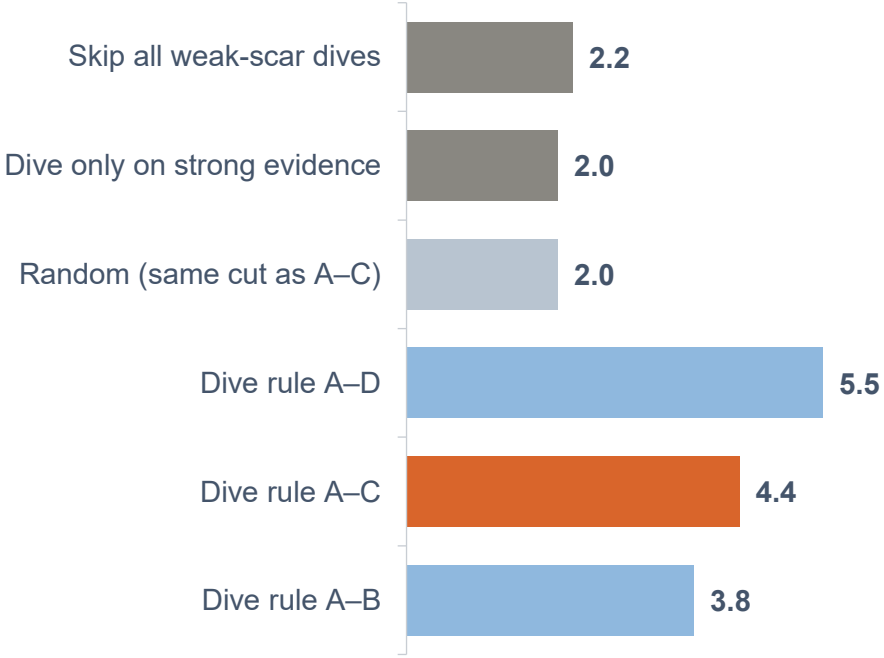
~1,400 a season; 22 more misses over 8 seasons

Same number of dives skipped at random: 24% of ghost dives avoided, 5.1% of dives above 0.04 missed (mean CPUE 0.128).

The ranking roughly **halves the misses** for the same cut... and the dives it does miss sit just above the threshold (median CPUE 0.069). COTS left behind are still there next visit.

Why not just skip weak evidence?

Ghost dives avoided per dive above 0.04 missed... the ranking earns its place in a targeted cut



Higher is better. A–C: 4.4 (95% CI 3.6–5.8).

Setting	Ghost avoided	> 0.04 missed
A–D cautious	18%	1.4%
A–C recommended	**30%**	**2.8%**
A–B aggressive	41%	4.5%

A–C is the sweet spot. Moving from A–D to A–C avoids 380 more ghost dives for 112 more misses... going on to A–B adds 379 for 136.

Field trial: one voyage, then shadow records

One voyage can test the tow zone... it can't test the dive rule (about 2 flagged dives per voyage)

1 One live voyage

Choose: a voyage with at least 3 follow-up reefs

Freeze: the ranking at departure, maps issued

Tow the zone first, then the rest... same ground, different order

Flag and defer: dive flagged sites last, so outcomes are still recorded

Record: tow times, overrides and a crew debrief

Go if maps arrive on time, crews apply the rules without disrupting the schedule, the zone saves time, and the rest of the reef finds no extra dives above 0.04.

2 A season of shadow records

Every voyage: freeze the ranking before departure

Change nothing on the water... crews do nothing new

Score afterwards from normal tow and dive records

~25 voyages (a third of a season) give a 95% chance of seeing the pattern

Adopt the dive rule if skipped dives have fewer above 0.04 and more ghosts than kept dives (backtest: 10% vs 22% above 0.04; 44% vs 32% ghost).

Take aways

- Write down **strong vs weak**... consistent triggers across crews
- Keep diving strong evidence; **rank only the rest**
- **~4 ghost dives avoided** per >0.04 dive missed... double random
- Tow zone saves **20% of follow-up tows**... needs a field test
- **The ask:** one trial voyage + a season of shadow records

